ANSWER SHEET DAC 2026

FINAL ROUND



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CHAPTER I: Introduction

In the life insurance business, long-term relationships with policyholders are the foundation of sustainable recurring premium income. Every policy that is discontinued not only removes future premium revenue, but also forces the company to spend more on acquiring new customers to maintain the size of its portfolio. For this reason, customer churn is one of the most important indicators that management needs to understand and control.

The company manages a diverse portfolio consisting of four insurance products, five premium bands, and five payment frequencies, held by customers with different profiles. However, churn does not occur evenly across the portfolio. Patterns that appear in one customer group may not apply to another, and not every pattern has the same business implication. Management therefore needs more than an overall churn rate, it needs to know where churn is concentrated, which factors are reliably associated with it, and what targeted actions can be taken.

The problem is formulated into four questions that follow the four challenges of the case: (1) what are the key patterns and characteristics of churn across the portfolio; (2) which parts of the portfolio warrant greater attention and offer meaningful retention opportunities; (3) which findings provide the strongest evidence of an association with churn and how strong is that evidence; (4) which retention strategies should the company consider and which policyholders should be the primary focus.

CHAPTER II: Theoretical Framework

A. Normality Test

Normality tests are used to decide whether parametric methods (which assume normally distributed data) or non-parametric methods are appropriate for the data. Various descriptive measures such as skewness, kurtosis, and the empirical distribution function have been proposed for testing normality, but only a few are frequently used in practice.

1. Shapiro-Wilk Test

   One of the most popular tests for normality with good power properties, based on the correlation between the ordered observations and the associated normal scores. The test statistic was derived by Shapiro and Wilk:

$$W = \frac{(\Sigma a_i y_{(i)})^2}{\Sigma (y - \underline{y})^2} \qquad (1)$$

   where $y_{(i)}$ is the i-th order statistics and a_i is the i-th expected value of normalized order statistics. It is most powerful for small samples and is commonly limited to $n \leq 5{,}000$.

2. Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test was first derived by Kolmogorov (1933) and later modified and proposed as a test by Smirnov (1948). The test statistic is

$$D = sup_x \, |F_n(X) - F(X, \mu, \sigma)| \qquad (2)$$

where $F(X, \mu, \sigma)$ is the theoretical cumulative distribution function of the normal distribution function and $F_n(X)$ is the empirical distribution function of the data. Large values of D indicate that the data are not normal. When μ and σ are unknown, sample estimates are used. This test is suitable for large samples.

B. Evaluation Metric

Evaluation metrics are quantitative measures used to assess the performance of a model or algorithm. They help determine how well a model is performing and guide improvements. The choice of evaluation metric depends on the specific task and goals of the model.

AUC-ROC curve is a graph used to check how well a binary classification model works. It helps us to understand how well the model separates the positive cases like people with a disease from the negative cases like people without the disease at different threshold level. It shows how good the model is at telling the difference between the two classes by plotting:

- **True Positive Rate (TPR):** How often the model correctly predicts the positive cases also known as Sensitivity or Recall.
- **False Positive Rate (FPR):** How often the model incorrectly predicts a negative case as positive.
- **Specificity:** Measures the proportion of actual negatives that the model correctly identifies. It is calculated as 1 - FPR.

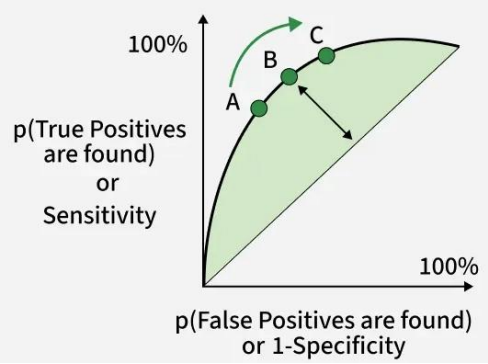


Figure 2.1 ROC AUC

C. Gradient Boosting Models

Gradient Boosting is a boosting algorithm and here each new model is trained to minimize the loss function such as mean squared error or cross-entropy of the previous model using gradient descent. In each iteration the algorithm computes the gradient of the loss function with respect to predictions and then trains a new weak model to predict this gradient. Predictions of the new model are then added to the ensemble (all models prediction) and the process is repeated until a stopping criterion is met.

Shrinkage and Model Complexityis a key feature of Gradient Boosting is shrinkage which scales the contribution of each new model using learning rate (denoted as η).

- **Smaller learning rates:** mean the contribution of each tree is smaller which reduces the risk of overfitting but requires more trees to achieve the same performance.
- **Larger learning rates:** mean each tree has a more significant impact but this can lead to overfitting.

There's a trade off between the learning rate and the number of estimators (trees) a smaller learning rate usually means more trees are required to achieve optimal performance.

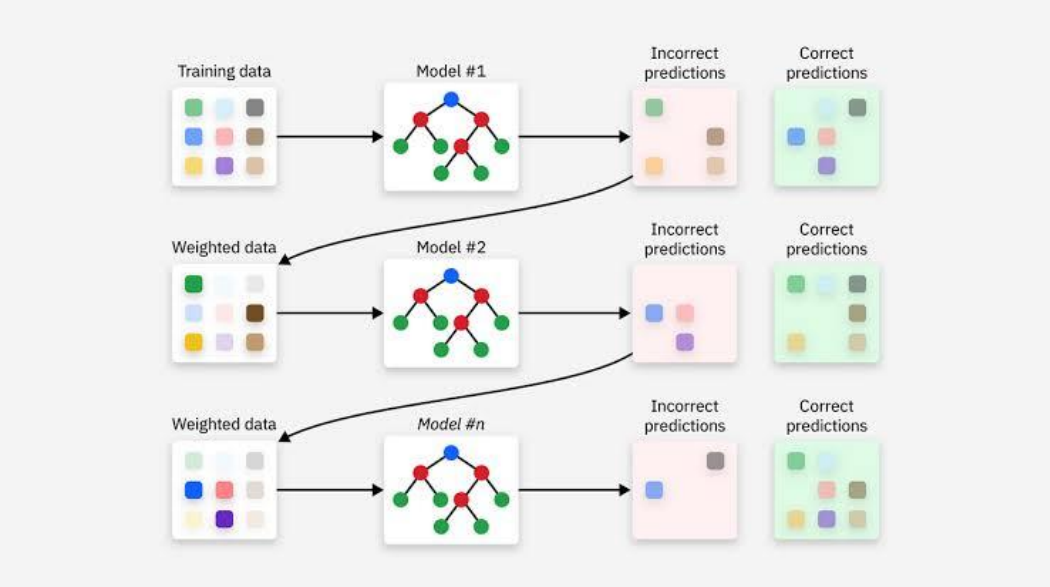


Figure 2.2 Gradient Boosting

CHAPTER III: Analytical Steps

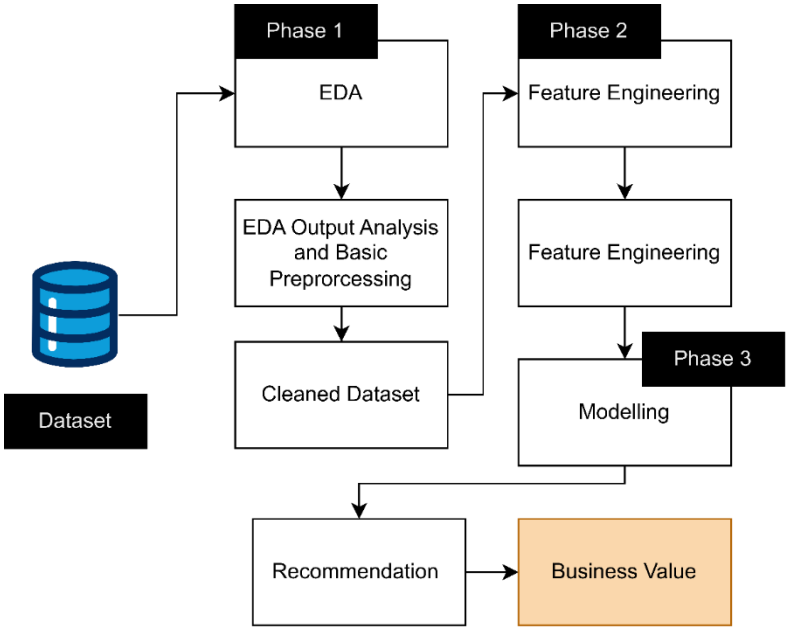


Figure 3.1 Workflow Diagram

In this chapter as seen in the Figure 3.1, we will analyze how we get the business value through several processes using the given datasets.

Phase 1 focuses on the Exploratory Data Analysis we will analyze the datasets and then outputting the cleaned dataset that is ready for the next phase that is phase 2. In phase 2 we will be doing Feature Engineering based on the phase 1 results. After the dataset is ready, the next phase is Modelling for developing a machine learning pipeline to predict the is_churn column for Recommendation and Business Value.

In Phase 1, we realized that the dataset file contained three tabs, policy, customer, and claim. Consequently, we converted each of these tabs into a separate CSV files. After that, these three files can be merged using *concat* method using pandas with customer_id as the superkey column or the connector between these 3 CSV files.

After the dataset has been merged, We processed the dataset by identifying missing values and calculating and processing categorical and numerical features. Additionally, we performed calculations to detect outliers within the numerical features. In addition, we also analyze the target values.

After receiving the EDA results, we proceeded to Phase 2, Feature Engineering to prepare features for the subsequent modeling process. We dropped the age, marital_status, and gender columns based on EDA results indicating that these features were weak in terms of their relationship with the target feature. In addition, we also carry out encoding for certain required columns.

For hold-out validation, the dataset is split into 80% training and 20% validation sets, whereas k-fold validation splits the dataset into 5 folds and repeatedly chooses 4 folds for training and 1 fold for validation. Both classical and ensemble models are trained on the training data for classification of target label. Ensemble models utilize its advantage where multiple models work together to make better predictions, whereas classical models rely on traditional feature processing. Then, the results of the predictions serve as a quantitative support tool in predicting a customer churn.

Next, we proceeded with modeling with ROC AUC as the evaluation metric using two approaches, first a baseline method employing logistic regression, and a final method utilizing three models (XGBoost, LightGBM, and CatBoost), which were then ensembled based on their respective performance. We then evaluated the results of this ensemble.

Subsequently, we conducted SHAP analysis and model interpretation to gain deeper insight into the model's predictions. Ultimately, the model's predictive results will prove highly valuable for forecasting customer churn rates in the future.

CHAPTER IV: Analysis of Results

As mentioned in the chapter before, the raw data need to be checked for missing values, duplication, and integrity. As a result, there are no duplicated rows or keys and every policy links to an existing customer. However, occupation is missing for 4,514 of 4,704 customers (95.96%), and the reported values are inconsistent free text (e.g., KARYAWAN SWASTA, KARIAWAN SWASTA, PEGAWAI SWASTA).

Therefore, occupation is not used as a category, only *occupation_reported* is kept. The claim table contains only 17 claims from 7 customers (8 of them from a single customer), so claim-related findings must be interpreted with great caution. Moreover, 195 customers hold more than one policy (up to 9), which means policies are not independent observations. After the basic preprocessing is complete, the dataset is ready to be used in each phase.

Before diving deep into Phase 1, Exploratory Data Analysis (EDA) serves as the critical initial step to understand the characteristics of the dataset. The first characteristic is the target distribution.

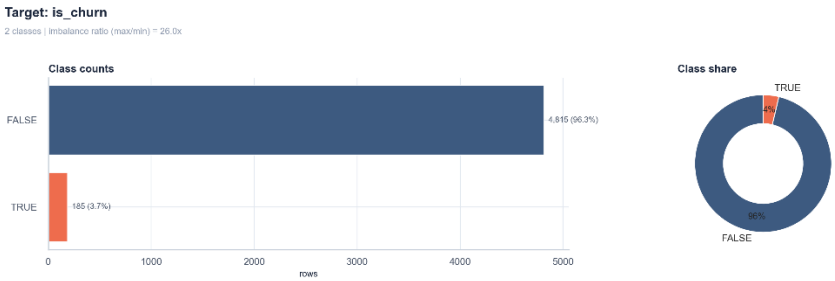


Figure 4.1. Distribution of the target variable is_churn

As shown in Fig. 4.1, only 185 of 5,000 policies (3.7%) churned, held by 121 customers (2.57% of customers). The data are highly imbalanced, so subsequent comparisons use churn rate and lift, and models are evaluated with ROC-AUC, PR-AUC, recall, and F1. Since the population size is large enough ($n > 50$), the experiment uses the Kolmogorov-Smirnov Test (supported by Shapiro-Wilk) with the null hypothesis (H0) that the data are normally distributed and the alternative hypothesis (H1) that the data are not normally distributed.

Next, we analyze the data feature by feature. As explained in the previous chapter, the features are divided into two categories, categorical and numerical. We can observe the distributions of both the categorical and numerical data.



Figure 4.2 Numerical and Categorical Distribution

After that, we can also carry out an analysis to identify the features that are most indicative of the target label, namely is_churn.

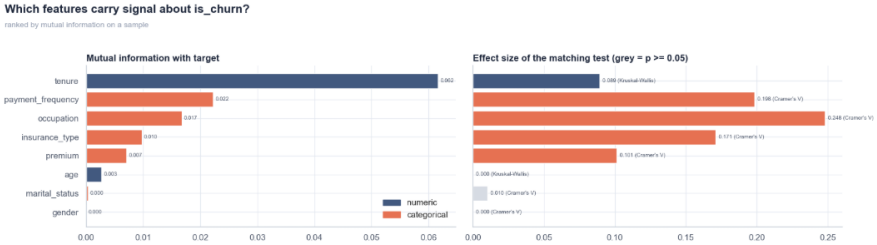


Figure 4.3 Indicative Features to Target

Here we can see that the tenure feature carries a much stronger signal than the other features; from this, we can conclude that the longer a person has been subscribed to an insurance policy, the more likely they are to churn. We can also see this in the graph below.

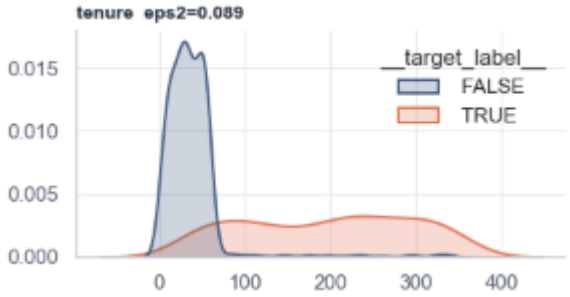


Figure 4.4 tenure to Target

With regard to the Categorical features section, we can see from the figure below that the two most influential features are payment_frequency and occupation_effect.

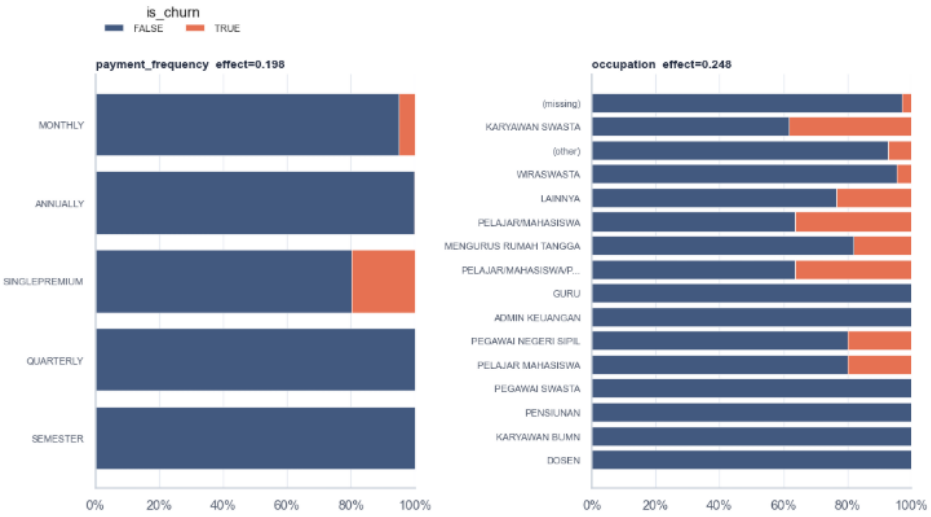


Figure 4.5 Categorical Indicative

From the final results of this EDA, we can conclude that the strongest feature-target association lies in:

- tenure (numeric): MI 0.062, Kruskal-Wallis stat 445.587, effect 0.089, p = 6.6e-99
- payment_frequency (categorical): MI 0.022, Cramer's V stat 0.198, effect 0.198, p = 2.7e-42
- occupation (categorical): MI 0.017, Cramer's V stat 0.248, effect 0.248, p = 1.7e-48

In addition, we can see the full correlation between categorical columns in:

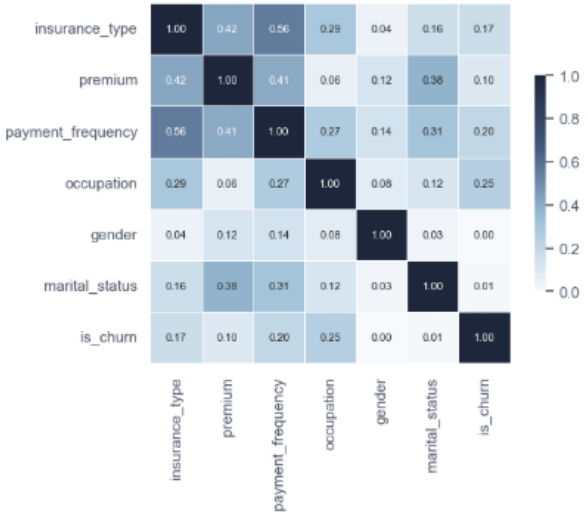


Figure 4.6 Categorical Correlation Diagram

For this Modelling section, as in the previous chapter, we use ROC AUC as the primary metric, employing a 5-fold validation strategy. For the baseline model, we use Logistic Regression as the primary model, with the following results:

```
[baseline] fold 1/5  roc_auc = 0.9847  (0s)
[baseline] fold 2/5  roc_auc = 0.9849  (0s)
[baseline] fold 3/5  roc_auc = 0.9805  (0s)
[baseline] fold 4/5  roc_auc = 0.9934  (1s)
[baseline] fold 5/5  roc_auc = 0.9879  (1s)
[baseline] CV roc_auc = 0.9857 +/- 0.0043  in 1s
```

	naive (CV)	baseline (CV)	naive (holdout)	baseline (holdout)
accuracy	0.9603	0.9686	0.9737	0.9717
balanced_accuracy	0.5000	0.9746	0.5000	0.9293
f1_macro	0.4899	0.8479	0.4933	0.8035
f1_weighted	0.9409	0.9726	0.9608	0.9758
logloss	0.1668	0.1402	0.1242	0.0995
roc_auc	0.5000	0.9857	0.5000	0.9873
pr_auc	0.0397	0.6367	0.0263	0.5570
f1	0.0000	0.7123	0.0000	0.6216

Figure 4.7 Baseline Results

The baseline results here are already quite high, with an accuracy of 0.9686 and an ROC AUC of 0.9857. We then trained three Gradient Boosting models and obtained the following results:

```
[lightgbm] fold 1/5  roc_auc = 0.9844  (0s)
[lightgbm] fold 2/5  roc_auc = 0.9876  (0s)
[lightgbm] fold 3/5  roc_auc = 0.9836  (1s)
[lightgbm] fold 4/5  roc_auc = 0.9923  (1s)
[lightgbm] fold 5/5  roc_auc = 0.9874  (1s)
[lightgbm] CV roc_auc = 0.9869 +/- 0.0031  in 1s
[xgboost] fold 1/5  roc_auc = 0.9837  (0s)
[xgboost] fold 2/5  roc_auc = 0.9881  (1s)
[xgboost] fold 3/5  roc_auc = 0.9825  (1s)
[xgboost] fold 4/5  roc_auc = 0.9923  (1s)
[xgboost] fold 5/5  roc_auc = 0.9872  (2s)
[xgboost] CV roc_auc = 0.9870 +/- 0.0035  in 2s
[catboost] fold 1/5  roc_auc = 0.9836  (4s)
[catboost] fold 2/5  roc_auc = 0.9854  (7s)
[catboost] fold 3/5  roc_auc = 0.9844  (66s)
[catboost] fold 4/5  roc_auc = 0.9938  (88s)
[catboost] fold 5/5  roc_auc = 0.9846  (98s)
[catboost] CV roc_auc = 0.9859 +/- 0.0037  in 98s
```

model	cv_roc_auc	cv_std	holdout_roc_auc	mean_best_iter	seconds
baseline	0.9857	0.0043	0.9873	NaN	0.7336
lightgbm	0.9869	0.0031	0.9903	332.8000	1.4876
xgboost	0.9870	0.0035	0.9888	239.0000	2.0249
catboost	0.9859	0.0037	0.9912	604.2000	97.9361

Figure 4.8 Gradient Boosting Results

Here we can see that the more advanced Gradient Boosting model achieves better results compared to the simple baseline model, namely logistic regression. We then created an ensemble using LightGBM (0.55), CatBoost (0.37) and XGBoost (0.08), and ultimately found that the ensemble model outperformed the other models.

model	cv_roc_auc	cv_std	holdout_roc_auc	seconds	cv_accuracy	cv_balanced_accuracy	cv_f1_macro	cv_f1_weighted	cv_logloss	cv_pr_auc	cv_f1
ensemble	0.9872	0.0036	0.9911	0.0000	0.9716	0.9611	0.8555	0.9747	0.0863	0.6350	0.7260
xgboost	0.9870	0.0035	0.9888	2.0249	0.9703	0.9514	0.8488	0.9736	0.0925	0.6488	0.7133
lightgbm	0.9869	0.0031	0.9903	1.4876	0.9718	0.9552	0.8551	0.9748	0.0906	0.6406	0.7251
catboost	0.9859	0.0037	0.9912	97.9361	0.9706	0.9696	0.8537	0.9741	0.0931	0.6092	0.7230
baseline	0.9857	0.0043	0.9873	0.7336	0.9686	0.9746	0.8479	0.9726	0.1402	0.6367	0.7123

```
Selected model: ensemble  (CV roc_auc = 0.9872)
```

Figure 4.9 Ensemble Models Results

Here, we can also analyse the final model results using the SHAP technique, where we can see that the results of our EDA—namely the ‘tenure’, ‘payment_frequency’ and ‘occupation_freq’ columns—have a significant influence on the model’s performance.

	gain_share	mean_abs_shap
tenure	0.6896	4.4489
payment_frequency	0.0724	1.0010
occupation_freq	0.1106	0.6726
occupation	0.0458	0.2934
payment_frequency_freq	0.0131	0.2554
insurance_type_freq	0.0269	0.2419
premium_freq	0.0138	0.1305
insurance_type	0.0083	0.1069
marital_status_freq	0.0065	0.0850
premium	0.0060	0.0743
gender_freq	0.0071	0.0735

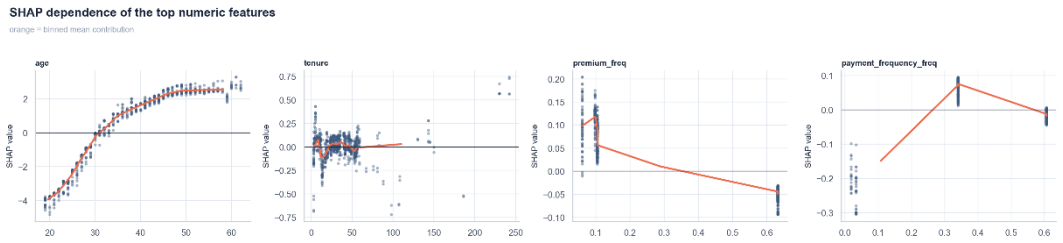


Figure 4.10 SHAP Analysis

Furthermore, we can also identify where the model fails.

	precision	recall	f1	support
TRUE	0.5330	0.9230	0.6760	26
FALSE	0.9980	0.9780	0.9880	964

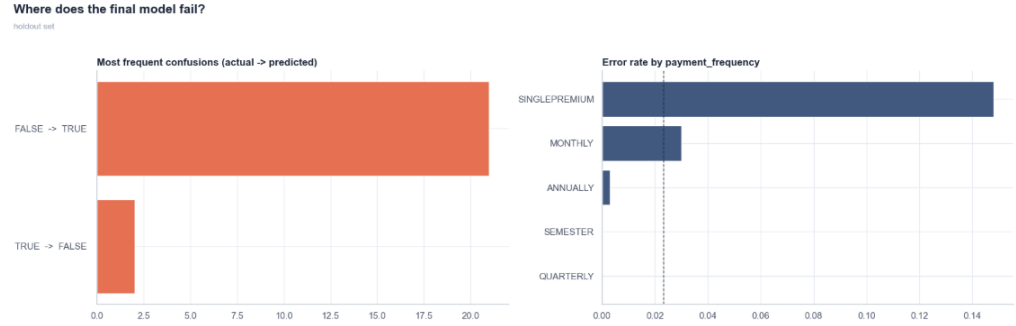


Figure 4.10 Model Fails

In the figure below, we can see that the model is overly biased towards ‘FALSE’ in its churn predictions; this is because the dataset distribution is heavily skewed towards ‘FALSE’. Consequently, additional data containing ‘is_churn’ values of ‘TRUE’ will be required in future to improve the model’s performance.

Dengan hasil analysis dan prediksi model ini kita bisa tau beberapa cara untuk mempertahankan customer agar tidak churn adalah dengan lebih memperhatikan customer yang sudah lama berlangganan sebuah asuransi, berikan insentif dan loyalty programs dan juga buat customer service menjadi lebih baik lagi. Selain itu tetap jaga pricing yang competitive agar customer baru tetap bertambah lebih banyak lagi

CHAPTER V: Conclusion and Recommendation

Based on our previous analysis, the key information used to predict customer churn is tenure, or the total duration of the policy contract prior to renewal. Certain factors, such as payment frequency and occupation, provide strong support for predictive modelling in customer churn classification. Based on the results of this analysis and the model’s predictions, we can identify several ways to retain customers and prevent churn, such as paying closer attention to long-standing policyholders, offering incentives and loyalty programmes, and further improving customer service. Furthermore, it is important to maintain competitive pricing to ensure that the number of new customers continues to grow.

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